

MANDATORY SYSTEM EXPLANATION DOCUMENT

Project: Grounded RAG Question Answering System

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1. Design Parameter & Selection Rationale

- **Parameter Chosen:**
Relevance Similarity Threshold (`SIMILARITY_THRESHOLD = 0.35` Cosine Similarity) paired with `CHUNK_SIZE = 500` characters (75 overlap).
- **Selection Rationale:**
sentence-transformers/all-MiniLM-L6-v2 produces L2-normalized 384-d vectors where inner product in FAISS (IndexFlatIP) equals exact Cosine Similarity in [-1.0, 1.0]. During empirical testing across sample documents, relevant in-domain queries produced similarity scores between 0.52 and 0.84, while out-of-domain queries (e.g. stock prices, celestial facts) yielded top scores below 0.28. Setting a threshold of 0.35 established a strict empirical boundary: relevant queries pass through for grounded answer generation, while unsupported queries immediately return 'The answer could not be found in the provided documents.' with 0 source chunks, preventing costly LLM calls and hallucinations.

2. Observed Failure & Root Cause

- **Observed Failure:**
PyMuPDF emitted deprecation warnings (warning: The fitz API is deprecated). Opening the server URL in Edge displayed `ERR_ADDRESS_INVALID`.
- **Root Cause Analysis:**
PyMuPDF deprecated its legacy 'fitz' module alias in v1.24+ in favor of direct 'import pymupdf'. The browser `ERR_ADDRESS_INVALID` error occurred because Web Browsers treat 0.0.0.0 as an invalid client navigation URL (it is a server-side wildcard listening setting), and FastAPI returned a 404 Not Found when navigating to root / without an explicit index route.
- **Resolution Applied:**
Refactored `app/ingestion.py` to use `import pymupdf` directly, added a root `GET /` route serving the Web UI, and supported both `GET` and `HEAD` HTTP pre-checks in `app/main.py`.

3. Tracked Metric & Empirical Insights

- **Metric Tracked:**
End-to-End Query Latency and Grounding Precision across a 10-question evaluation benchmark (`evaluation/evaluate.py`).
- **Empirical Insights:**
Benchmark testing revealed an Average Query Latency of 57.38 ms and a Grounding Accuracy of 100.0% (10/10 correct). This data proved that coupling local dense vector embeddings (all-MiniLM-L6-v2) with in-memory FAISS indexing achieves sub-100ms retrieval speed while maintaining zero hallucination rates on out-of-domain questions.

4. Items Not Finished & Future Roadmap

- **Items Not Finished:**
1) Scanned PDF OCR: Image-only PDFs lacking a text layer return a 400 validation error rather than running OCR. 2) Vector Store Persistence: The FAISS index is maintained in-memory for MVP speed and resets on server restart.
- **Next Execution Steps:**
1) Implement `faiss.write_index / faiss.read_index` to persist vector indices to disk across server restarts. 2) Add Tesseract OCR (pytesseract) for scanned image PDF text extraction. 3) Integrate a Cross-Encoder reranker (ms-marco-MiniLM-L-6-v2) to re-score Top-5 candidates for complex queries.